

Notes: Grossman and Helpman (RES, 2005)

Outsourcing in a Global Economy

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Model environment	4
3.1	Countries, goods, and preferences	4
3.2	Final-good producers and input requirements	4
3.3	Component suppliers	5
3.4	Demand, pricing, and surplus inside a successful relationship	5
4	Benchmark: unverifiable relationship-specific investment	6
4.1	When does a match become viable?	6
4.2	Final producer search profits	6
4.3	Component supplier profits	6
5	Equilibrium with binding investment constraints	7
5.1	Equal-profit condition	7
5.2	Reduced-form SS and NN curves	7
5.3	Multiple equilibria and stability	8
6	Comparative statics without partial contracts	8
6.1	Growth in the south	8
6.2	Growth in the north	8
6.3	Improvements in customization technology	9
7	Partial verifiability and the contracting environment	9
7.1	Why partial contracts matter	9
7.2	Investment contracts	10
7.3	Why better contracting can have non-obvious aggregate effects	10
8	Comparative statics with partial contracts	11
8.1	Improving contracting in the north	11

8.2	Improving contracting worldwide	11
8.3	Improving contracting in the south	12
9	Extracted figures from the paper	13
9.1	Figure 1: Equilibrium curves: binding investment regimes and Figure 2: Equilibria with binding investment constraints in north and south	13
9.2	Figure 3: Labour supply growth in the south and Figure 4: Technological improvement in the south	14
9.3	Figure 5: Contracting improves in the north: low initial γ^N and Figure 6: Contracting improves worldwide	15
10	Main results in compact form	15
11	How to use the paper	16
11.1	For international trade	16
11.2	For organizational economics	16
11.3	For development and policy	16
11.4	For interpreting globalization	16
12	Limitations and assumptions	16
13	Short summary	17
14	Conclusion	17
14.1	Core conclusion	17
14.2	Economic intuition	17
14.3	Incremental contribution revisited	18
14.4	Policy implication	18
14.5	Final takeaway	18

1 Big picture

- **Question.** What determines whether a northern final-good producer outsources a specialized input at home or abroad?
- **Core idea.** Outsourcing is not modeled as buying a standardized input in a spot market. It requires search for a suitable supplier, a bilateral relationship, and a relationship-specific investment by the supplier.
- **Central trade-off.** The south may have lower wages, but the north may have a thicker supplier market, better search conditions, better customization technology, or a better contracting environment. A low wage is only one force among several.
- **Main mechanism.** The model creates a *thick-market externality*: final producers want to search where many suppliers are active, and suppliers want to enter where many final producers search.
- **Main equilibrium implication.** Outsourcing locations can be self-reinforcing. A country can attract more outsourcing not only because it is intrinsically cheaper, but also because its supplier market is already thick.
- **Main comparative statics.** Southern market growth and southern improvements in customization technology tend to move outsourcing toward the south. By contrast, improvements in contracting possibilities can have subtle and sometimes counterintuitive effects because they also change product-market entry and labor demand.
- **Paper type.** This is a theoretical general-equilibrium model of global outsourcing. The paper contains six formal figures and no formal tables.

Economic takeaway: Global outsourcing is jointly determined by wages, market thickness, search costs, customization costs, and contractibility. The model explains why a low-wage country may fail to attract outsourcing if its supplier market is thin, and why improvements in institutions do not always move outsourcing in the most intuitive direction.

2 Incremental contribution of the paper

- **From standard input trade to relationship-based outsourcing.** The paper treats outsourcing as a relationship-specific activity, not as anonymous trade in intermediate goods.
- **Search and matching in international sourcing.** Final producers must choose a country in which to search for a partner. This introduces market thickness as an endogenous determinant of sourcing location.
- **Incomplete contracts in a trade model.** The supplier must customize inputs before production. Because investment is partly nonverifiable, hold-up affects whether a match becomes viable.
- **General-equilibrium feedback.** Outsourcing decisions affect labor demand, relative wages, product entry, and trade volumes. These feedbacks are essential for the paper's counterintuitive comparative statics.

- **Multiple equilibria from a thick-market externality.** A country may have many suppliers because many final producers search there, and many final producers may search there because many suppliers are present.
- **Location rather than ownership.** Grossman and Helpman (2002a) study integration versus outsourcing. This paper holds outsourcing fixed and studies where outsourcing takes place.
- **Institutional nuance.** Better contracting in a country raises the private attractiveness of searching there, but can also change labor allocation and final-good entry in ways that may reduce international outsourcing.
- **Bridge to fragmentation/global production sharing.** The paper provides a formal explanation for why advances in technology, legal institutions, and market size can reshape global value chains.

3 Model environment

3.1 Countries, goods, and preferences

There are two countries: **north** (N) and **south** (S). There are two sectors.

- A homogeneous good z can be produced in either country with one unit of local labor per unit of output.
- Differentiated final goods y can be designed and assembled only in the north.
- Both countries can produce specialized intermediate inputs, called components.

Consumers have Cobb-Douglas preferences over the homogeneous good and a CES aggregate of differentiated varieties:

$$u = z^{1-\beta} \left[\int_0^1 \int_0^{\hat{n}(l)} y(j, l)^\alpha dj dl \right]^{\beta/\alpha}, \quad 0 < \alpha, \beta < 1.$$

The elasticity of substitution across differentiated varieties is

$$\varepsilon = \frac{1}{1 - \alpha}.$$

Interpretation: Demand is standard monopolistic-competition demand, but the input side is nonstandard. Each final-good variety needs a customized component, and this customization is the source of the outsourcing problem.

3.2 Final-good producers and input requirements

A northern final-good producer must first pay a fixed product-design cost:

$$w^N f_n.$$

The final-good design determines the location of the required input on a unit circle. The circle indexes input specifications. A point on the circle is not a geography; it is a technology or product-design location.

After designing a product, the final producer must search for a component supplier either in the north or in the south. Search in country $i \in \{N, S\}$ reveals the supplier in that country whose expertise is closest to the final producer's required input.

Interpretation: The final-good firm does not know *ex ante* which country contains a close technological match. Search is country-specific, so the firm must decide where to look.

3.3 Component suppliers

A component supplier enters in country i by paying

$$w^i f_m^i.$$

Each supplier develops expertise at a location on the unit circle. In equilibrium, suppliers in a country space themselves evenly around the circle. Let m^i denote the number of suppliers in country i .

If a final producer's required component is at distance x from a supplier's expertise, the supplier must invest in a prototype at cost

$$w^i \mu^i x.$$

Here μ^i measures the cost of customizing inputs in country i .

Interpretation: A thicker market, meaning higher m^i , makes it more likely that a final producer finds a nearby supplier. A lower μ^i makes it cheaper for a supplier to customize a component for a distant final producer.

3.4 Demand, pricing, and surplus inside a successful relationship

The demand for a final variety assembled with components from country i is

$$y^i = A \left(\frac{w^i}{\alpha} \right)^{-\varepsilon},$$

with the monopoly price

$$p^i = \frac{w^i}{\alpha}.$$

The joint operating surplus generated by a successful final-producer/component-supplier relationship is

$$S^i = (1 - \alpha) A \left(\frac{w^i}{\alpha} \right)^{1-\varepsilon}.$$

After the prototype is built, the relationship is efficient *ex post* and the parties Nash-bargain over the operating surplus. In the benchmark case, they split S^i equally.

Economic intuition: The contracting friction is not in the *ex post* production decision. It is in the *ex ante* prototype investment: the supplier must invest before the final producer can order the customized input.

4 Benchmark: unverifiable relationship-specific investment

4.1 When does a match become viable?

In the benchmark model, the supplier's customization investment is not verifiable. Therefore, the supplier will invest only if its later bargaining share covers the investment cost:

$$\frac{S^i}{2} \geq w^i \mu^i x.$$

Thus the greatest distance at which a successful relationship can form is

$$r^i = \min \left\{ \frac{S^i}{2w^i \mu^i}, \frac{1}{2m^i} \right\}.$$

The second term, $1/(2m^i)$, is the maximum possible distance to the nearest supplier when suppliers are evenly spaced. The first term is the incentive constraint created by the supplier's investment problem.

Interpretation: A final producer may find a technically closest supplier, but that does not mean outsourcing succeeds. If the supplier is too far from the required component specification, the relationship-specific investment is too costly relative to the supplier's expected share of surplus.

4.2 Final producer search profits

If a final producer searches in country i , the probability of finding a supplier within viable distance r^i is $2r^i m^i$. Conditional on a viable match, the final producer receives half of the operating surplus. Expected operating profits from searching in country i are therefore

$$\pi_n^i = r^i m^i S^i.$$

The final producer chooses the country with the higher π_n^i . If both domestic and foreign outsourcing occur in equilibrium, then

$$\pi_n^N = \pi_n^S.$$

Free entry into final-good design gives

$$\max\{\pi_n^N, \pi_n^S\} = w^N f, \quad f = f_n + f_s.$$

Interpretation: The final producer chooses a search market by comparing expected match quality, expected surplus, and the probability that the supplier will actually invest.

4.3 Component supplier profits

A supplier in country i serves the final producers whose input requirements lie within distance r^i on either side of its expertise. If a final producer is at distance x , the supplier earns

$$\frac{S^i}{2} - w^i \mu^i x.$$

Operating profit for an input supplier is

$$\pi_m^i = 2n^i \int_0^{r^i} \left(\frac{S^i}{2} - w^i \mu^i x \right) dx = r^i n^i (S^i - w^i \mu^i r^i).$$

Free entry gives

$$\pi_m^i \leq w^i f_m^i, \quad (\pi_m^i - w^i f_m^i) m^i = 0.$$

Economic intuition: More final producers searching in a country makes entry more attractive for suppliers. More suppliers in a country makes search more attractive for final producers. This two-sided feedback is the thick-market externality.

5 Equilibrium with binding investment constraints

The paper focuses much of the analysis on stable equilibria where outsourcing occurs in both countries and where the investment constraint binds in both countries. In that case,

$$r^i = \frac{S^i}{2w^i \mu^i}.$$

5.1 Equal-profit condition

Let

$$\omega = \frac{w^N}{w^S}$$

be the northern relative wage. When both countries host outsourcing, final producers must be indifferent between searching in north and south. In the binding-investment regime, this gives

$$\omega = \left(\frac{\mu^S m^N}{\mu^N m^S} \right)^{\frac{1-\alpha}{1+\alpha}}.$$

Interpretation: Search in the south becomes more attractive when the southern supplier market is thicker (m^S high) or southern customization is cheaper (μ^S low). To keep firms indifferent, the wage ratio must adjust.

5.2 Reduced-form SS and NN curves

Combining labor-market clearing with the equal-profit condition gives two reduced-form curves in (m^S, m^N) space:

$$\begin{aligned} SS : \quad m^N &= \frac{\mu^N}{\mu^S} m^S \left[\frac{\beta L^S - 2 \frac{1+\alpha}{1-\alpha} m^S f_m^S}{(1-\beta) L^N} \right]^{\frac{1+\alpha}{1-\alpha}}, \\ NN : \quad m^S &= \frac{\mu^S}{\mu^N} m^N \left\{ \frac{\left[1 - \frac{1}{2}(1-\alpha)\beta \right] L^N - 2 \frac{1+\alpha}{1-\alpha} m^N f_m^N}{\frac{1}{2}(1-\alpha)\beta L^S} \right\}^{\frac{1+\alpha}{1-\alpha}}. \end{aligned}$$

The SS curve describes combinations of northern and southern supplier counts that clear the southern labor market and keep search equally profitable. The NN curve does the same for the northern labor market.

Interpretation: The curves are not simple supply and demand curves. They combine labor clearing, wage adjustment, search profitability, supplier entry, and final-good entry.

5.3 Multiple equilibria and stability

Because of the thick-market externality, multiple equilibria can arise. One stable equilibrium can feature outsourcing in both countries. Another stable equilibrium can feature all component production concentrated in the north.

A useful way to think about stability is:

- below the SS curve, southern suppliers earn positive profits and enter;
- above the SS curve, southern suppliers exit;
- to the left of the NN curve, northern suppliers enter;
- to the right of the NN curve, northern suppliers exit.

Economic message: Location patterns in outsourcing can be path-dependent. A supplier market can survive because it is already thick, even if another location might also support outsourcing under different historical conditions.

6 Comparative statics without partial contracts

6.1 Growth in the south

An increase in the southern labor force L^S shifts the SS curve upward and the NN curve leftward. In the stable equilibrium with outsourcing in both countries:

- the number of southern component suppliers m^S rises;
- the number of northern component suppliers m^N falls;
- the relative wage of the south rises, meaning $\omega = w^N/w^S$ falls;
- international outsourcing expands;
- the ratio of trade to world income rises;
- the share of intra-industry trade rises.

Interpretation: A larger southern economy has a thicker potential supplier market and a larger labor pool. The market-thickness effect pulls more search and supplier entry into the south.

6.2 Growth in the north

An increase in northern labor supply has the opposite qualitative effect:

- northern component production expands;
- southern component production contracts;
- international outsourcing falls relative to domestic outsourcing;
- the south's relative wage falls;
- trade shares decline relative to the case of southern growth.

Economic message: Country size matters not only through final demand, but also through the thickness of supplier markets and the general-equilibrium wage response.

6.3 Improvements in customization technology

If customization technology improves proportionally in both countries, so both μ^N and μ^S fall by the same percentage, there is no effect on the location of outsourcing in the benchmark regime. Only the ratio μ^N/μ^S matters.

If instead customization technology improves disproportionately in the south, so μ^S falls relative to μ^N , then:

- searching in the south becomes more profitable at the initial wage ratio;
- the SS curve shifts upward and the NN curve shifts leftward;
- more component suppliers enter in the south;
- component suppliers exit in the north;
- outsourcing shifts from north to south;
- the ratio of trade to world income and the share of intra-industry trade rise.

Interpretation: The technological catch-up of southern suppliers can generate the kind of rising international outsourcing associated with global production sharing.

7 Partial verifiability and the contracting environment

7.1 Why partial contracts matter

The benchmark assumes the supplier's customization investment is completely unverifiable. The paper then allows a fraction $\gamma^i < 1/2$ of the investment in country i to be verifiable by an outside party. This parameter summarizes the quality of the legal and contracting environment.

- Higher γ^i means more of the supplier's investment can be governed by contract.
- The parties can write an investment contract before the order contract.
- The investment contract can specify a verifiable partial investment and an up-front payment.

Interpretation: Better legal institutions do not eliminate incomplete contracts, but they enlarge the set of relationships in which a supplier is willing to customize the input.

7.2 Investment contracts

Let a final producer and supplier in country i be separated by distance x in input space.

There are three cases:

1. If

$$\frac{S^i}{2} < (1 - \gamma^i)w^i\mu^i x,$$

the supplier's expected ex post share does not cover the noncontractible part of the investment.
No viable investment occurs.

2. If

$$\frac{S^i}{2} \geq w^i\mu^i x,$$

the supplier would invest even without an investment contract.

3. If

$$(1 - \gamma^i)w^i\mu^i x \leq \frac{S^i}{2} < w^i\mu^i x,$$

an investment contract allows the parties to share the cost of the verifiable part and make the full relationship-specific investment viable.

The payment in the investment contract is

$$P^i(x) = \begin{cases} \frac{1}{2}w^i\mu^i x, & \text{if } \frac{S^i}{2w^i\mu^i} < x \leq \frac{S^i}{2w^i\mu^i(1-\gamma^i)}, \\ 0, & \text{otherwise.} \end{cases}$$

The induced investment is

$$I^i(x) = \begin{cases} w^i\mu^i x, & \text{if } x \leq \frac{S^i}{2w^i\mu^i(1-\gamma^i)}, \\ 0, & \text{otherwise.} \end{cases}$$

The maximum viable distance becomes

$$r^i = \min \left\{ \frac{S^i}{2w^i\mu^i(1-\gamma^i)}, \frac{1}{2m^i} \right\}.$$

Economic message: A better legal environment raises the probability that a given search will yield a viable match because it expands the set of supplier-final-producer distances for which customization occurs.

7.3 Why better contracting can have non-obvious aggregate effects

A higher γ^i has a direct partial-equilibrium effect: it makes outsourcing in country i more attractive. But in general equilibrium it also changes:

- how many final-good producers enter;
- how much labor is used by component suppliers;

- relative wages;
- demand for the homogeneous good;
- the profitability of search in the other country.

This is why improvements in contracting can shift outsourcing in unexpected directions.

8 Comparative statics with partial contracts

8.1 Improving contracting in the north

Starting from $\gamma^N = \gamma^S = 0$, a small increase in γ^N makes searching in the north more profitable. To restore equal search profitability, the northern relative wage must rise.

In the stable equilibrium:

- m^N rises;
- m^S falls;
- domestic outsourcing in the north rises;
- international outsourcing to the south falls;
- the ratio of trade to world income falls;
- the intra-industry trade share falls.

However, the effect is not globally monotone. With larger initial γ^N , further improvements in northern contracting can eventually produce opposite responses through labor-demand and entry effects.

Economic interpretation: Better northern contracts directly raise the attractiveness of northern suppliers. But because they also change how many final producers enter and how much labor component suppliers use, the response can become nonmonotone.

8.2 Improving contracting worldwide

Suppose $\gamma^N = \gamma^S = 0$ initially and both countries experience the same improvement in contractibility. This might look neutral because the relative contracting environment does not change. But the paper shows it is not neutral.

A worldwide improvement in contracting tends to shift outsourcing from the south to the north:

- better contracts raise the probability that a search becomes a viable relationship;
- fewer final-good producers need to enter to satisfy product demand;
- resources are freed in the north from final-good design/search activities;
- the north can expand component supply;

- higher world income raises demand for the homogeneous good, pulling southern labor toward that sector;
- fewer southern workers remain available for component production.

Economic message: Even symmetric legal improvements can have asymmetric location effects because the north is the only country that designs and assembles final differentiated goods.

8.3 Improving contracting in the south

If southern contracting improves from a low level while northern contracting is already better, the direct effect is to make searching in the south more attractive. But the general-equilibrium effects are ambiguous.

The paper emphasizes that:

- the direct relative-profitability effect pushes outsourcing toward the south;
- southern component producers may then demand more labor for investment and production;
- stronger competition among final producers can reduce northern labor demand;
- both SS and NN curves can shift in either direction;
- numerical results often show southern outsourcing rising at first and then falling below its original level.

Economic message: The largest amount of international outsourcing may occur when the south has somewhat less complete contracts than the north, not when the south has fully caught up institutionally.

9 Extracted figures from the paper

9.1 Figure 1: Equilibrium curves: binding investment regimes and Figure 2: Equilibria with binding investment constraints in north and south

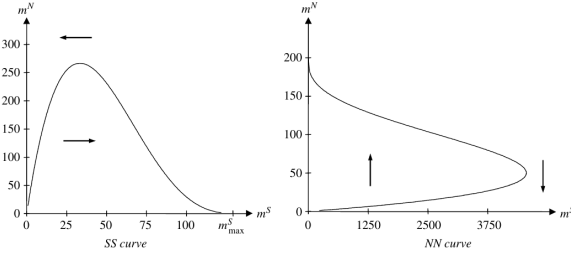


FIGURE 1
Equilibrium curves: binding investment regimes

Figure 1: Equilibrium curves: binding investment regimes

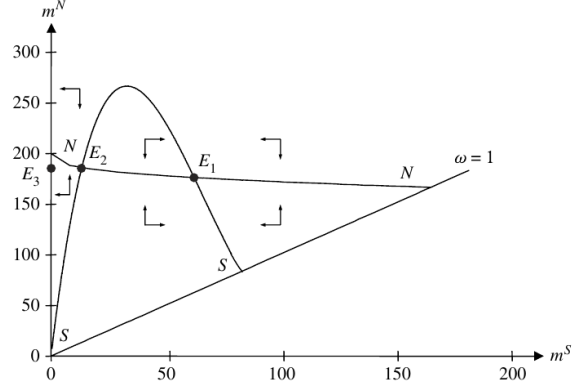


FIGURE 2
Equilibria with binding investment constraints in north and south

Figure 2: Equilibria with binding investment constraints in north and south

Read: Figure 1 plots the SS and NN equilibrium loci separately. The SS curve combines southern labor-market clearing with equal search profitability; the NN curve does the analogous job for the north. Figure 2 overlays the two curves and adds the ray $\omega = 1$.

Economic message: The model can generate several equilibria because the outsourcing location decision is self-reinforcing. Point E_1 represents a stable equilibrium with outsourcing in both countries. Point E_2 is unstable. Point E_3 represents a stable equilibrium in which outsourcing is concentrated in the north. The figures make the thick-market externality visible: suppliers enter where final producers search, and final producers search where suppliers are dense.

9.2 Figure 3: Labour supply growth in the south and Figure 4: Technological improvement in the south

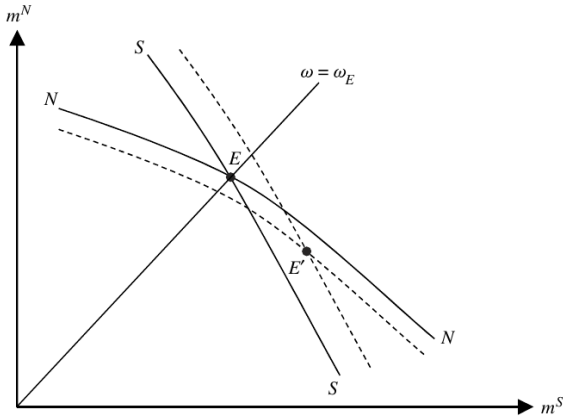


FIGURE 3
Labour supply growth in the south

Figure 3: Labour supply growth in the south

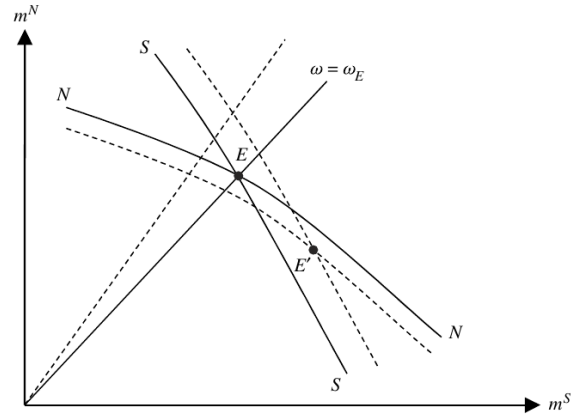


FIGURE 4
Technological improvement in the south

Figure 4: Technological improvement in the south

Read: Figure 3 shows the effect of an increase in southern labor supply. The stable equilibrium shifts from E to E' , with more southern component suppliers and fewer northern component suppliers. Figure 4 shows a similar qualitative movement after a southern improvement in customization technology.

Economic message: Southern growth and southern technological catch-up both make the south a more attractive outsourcing location. They thicken the southern supplier market and shift outsourcing activity from the north to the south. In the model, these changes also increase the trade share of world income and raise the share of intra-industry trade.

9.3 Figure 5: Contracting improves in the north: low initial γ^N and Figure 6: Contracting improves worldwide

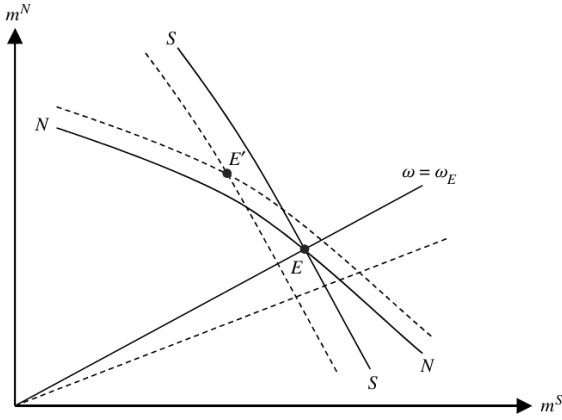


FIGURE 5
Contracting improves in the north: low initial γ^N

Figure 5: Contracting improves in the north: low initial γ^N

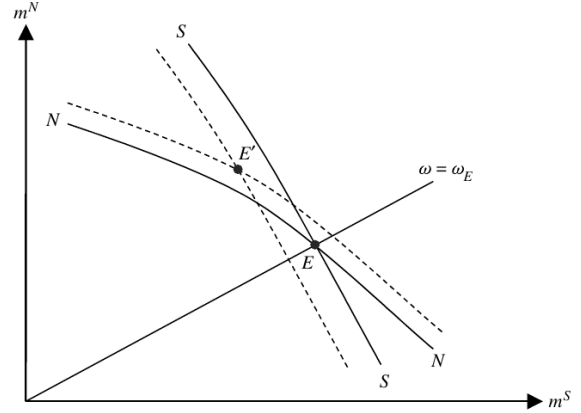


FIGURE 6
Contracting improves worldwide

Figure 6: Contracting improves worldwide

Read: Figure 5 shows that an initial improvement in northern contractibility shifts the equilibrium toward more northern component suppliers and fewer southern suppliers. Figure 6 shows that even a worldwide improvement in contracting possibilities can shift outsourcing toward the north.

Economic message: Contracting improvements are not equivalent to wage reductions or technology improvements. They alter the viability of matches, the entry of final-good producers, and the allocation of labor between sectors. Because final-good design and assembly occur only in the north, better contracting can free northern resources for component production and reduce international outsourcing.

10 Main results in compact form

1. **Outsourcing is relationship-specific.** A successful outsourcing relationship requires a supplier close enough in input space and willing to invest in customization.
2. **Market thickness matters.** A final producer prefers to search in a country with many suppliers because the closest match is likely to be better.
3. **Supplier entry and final-producer search are complements.** More suppliers attract more search; more search attracts more suppliers.
4. **Multiple equilibria are possible.** A location can remain an outsourcing hub because it already has a thick supplier market.
5. **Southern growth expands international outsourcing.** A larger south generally attracts more component production and increases trade intensity.

6. **Symmetric technology improvements are neutral.** If both countries improve customization technology proportionally, the location of outsourcing is unchanged.
7. **Southern technological catch-up shifts outsourcing south.** A fall in μ^S relative to μ^N makes southern search more attractive and expands southern outsourcing.
8. **Better contracts expand the set of viable matches.** A higher γ^i lets parties support relationships that would fail under complete nonverifiability.
9. **Contracting effects are subtle in general equilibrium.** Better contracting in the north can shift outsourcing north; symmetric worldwide improvements can also shift outsourcing north; better contracting in the south can have ambiguous effects.

11 How to use the paper

11.1 For international trade

The paper provides a theory of fragmentation that goes beyond Ricardian productivity and wage differences. It explains why global value chains depend on supplier-market thickness, contracting institutions, and search frictions.

11.2 For organizational economics

The paper embeds incomplete contracts and relationship-specific investment into a trade model. It shows how micro-level hold-up frictions can shape macro-level trade patterns.

11.3 For development and policy

The model suggests that low wages are not enough for a country to become an outsourcing destination. A country also needs a sufficiently thick supplier base, good search infrastructure, capable customization technology, and credible contracting institutions.

11.4 For interpreting globalization

The model helps explain why outsourcing may cluster in some locations, why production sharing can grow rapidly after market-size or technology changes, and why institutional reform can have nonmonotone effects on trade.

12 Limitations and assumptions

- **No make-or-buy choice.** The paper assumes final producers must outsource the component. It does not simultaneously model vertical integration versus outsourcing.
- **Final-good design only in the north.** This asymmetry is central to some results, especially the effect of worldwide contracting improvements.

- **Simplified search.** Search is country-specific and reveals the closest supplier. More complex search intensity choices are discussed in related work, but the published model keeps search simple.
- **Even supplier spacing.** Suppliers locate evenly around the input-space circle. This gives tractable market-thickness effects.
- **No empirical test.** The paper is theoretical. Its contribution is conceptual and comparative-static, not econometric.
- **Stable-equilibrium selection.** The model can produce multiple equilibria, so comparative statics are interpreted around stable equilibria.

Interpretation: These simplifications are deliberate. They allow the model to isolate the interaction between search, incomplete contracts, and general equilibrium wages.

13 Short summary

Grossman and Helpman develop a general-equilibrium model of global outsourcing in which final-good producers in the north search for component suppliers either at home or abroad. Outsourcing succeeds only if the selected supplier is close enough in input space and willing to make a relationship-specific investment. Because more suppliers make search more attractive and more searching final producers make supplier entry more attractive, the model features a thick-market externality and possible multiple equilibria.

The paper shows that southern growth and southern improvements in customization technology tend to move outsourcing toward the south. However, contracting improvements are more subtle. Better contracts in a country directly make outsourcing there more attractive, but they also affect final-good entry, labor demand, wages, and the allocation of labor across sectors. As a result, worldwide improvements in contracting can shift outsourcing toward the north, and improvements in the south can have nonmonotone effects.

14 Conclusion

14.1 Core conclusion

The location of outsourcing is determined by much more than relative wages. It depends on the thickness of supplier markets, the cost of search, the cost of customization, and the contracting environment. These forces interact in general equilibrium.

14.2 Economic intuition

A country becomes an attractive outsourcing location when final producers expect to find a supplier who is both technologically close and willing to invest. Supplier density, legal enforceability, and customization technology all raise the expected return from searching in that country.

14.3 Incremental contribution revisited

The paper's distinctive contribution is to combine three ideas: search for partners, relationship-specific investment under incomplete contracts, and international general equilibrium. This combination generates insights that would be missed in a model of anonymous input trade or a partial-equilibrium model of contracting.

14.4 Policy implication

For a country trying to attract outsourcing, low wages are insufficient. Policies that deepen supplier markets, reduce search costs, improve technical capability, and strengthen contracting institutions can matter. But the paper also warns that institutional reforms can have complex equilibrium effects, especially when final-good production is geographically asymmetric.

14.5 Final takeaway

Outsourcing is a market-design and institution problem as much as a wage-cost problem. The geography of global production depends on where firms can find, contract with, and rely on specialized partners.